Low-Power mmWave Radar Sensors for Atmospheric Boundary Layer Characterization

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ABSTRACT

Highly integrated frequency-modulated continuous-wave (FMCW) millimeter-wave (mmWave) radars provide a low-cost path toward distributed atmospheric sensing, but environmental targets often produce weak backscatter compared to receiver noise and ground clutter. This project employs an 77-81 GHz Texas Instruments (TI) mmWave radar evaluation module for atmospheric boundary-layer observations by combining **(i)** DSP/firmware-level configuration studies to adapt the radar's processing chain and data-cube formation for atmospheric returns, **(ii)** custom post-DSP processing to reduce noise and enhance weak-target detectability, **(iii)** systematic evaluation of radar configuration trade-offs impacting signal-to-noise ratio (SNR) and range/velocity estimation, and **(iv)** a deployable external control platform enabling long-duration, unattended operation in remote, outdoor environments.

A Python TLV (Type-Length-Value) parser was developed to extract DSP outputs (range–Doppler products) for flexible averaging and analysis. Frame averaging reduces noise-floor variance and improves visibility of weak returns, with quantified impacts on effective frame rate. The hardware and software structure enables power sequencing, data integrity, fault recovery, and secure remote monitoring. Indoor and controlled outdoor tests confirm pipeline correctness and demonstrate improved stability and weak-return sensitivity, advancing field readiness for future precipitation detection and wind-profiling studies.



Fig. 1: mmWave radar system deployed as part of the Meteorological Sensor Array at the Jornada Experimental Range, NM as of June 2, 2026.

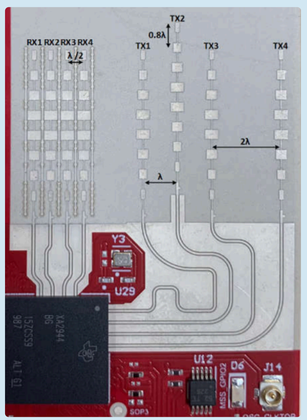


Fig. 2: TI mmWave AWR2944 FMCW SoC radar with 4 TX, 4 RX antenna array, operating in the 77-81 GHz band

GOAL

Repurpose a 77-81 GHz TI automotive mmWave radar (AWR) into a high-resolution atmospheric sensor by optimizing DSP/firmware settings, applying SNR-enhancing processing, and engineering a robust remote deployment platform for long-duration experiments.

BACKGROUND INFO

Why mmWave radar for the atmosphere?

- Boundary-layer processes (precipitation, turbulence, winds) evolve quickly and vary spatially → need dense, high-cadence observations
- Conventional radar systems are high-performance but typically high-cost, sparse, and affected by blockage
- Automotive mmWave FMCW radar offers:
  - Small, low power, low cost
  - Configurable waveforms and high range resolution
  - Potential for distributed networks (many nodes)

Core challenge for atmospheric sensing

- Hydrometeors and boundary-layer targets often return weak signals.
- Returns are frequently buried under:
  - Receiver noise (thermal + system noise)
  - Stationary clutter (ground/structures)
- Improving SNR and operational reliability is essential to extend detection range and enable long unattended deployments.

SYSTEM OVERVIEW

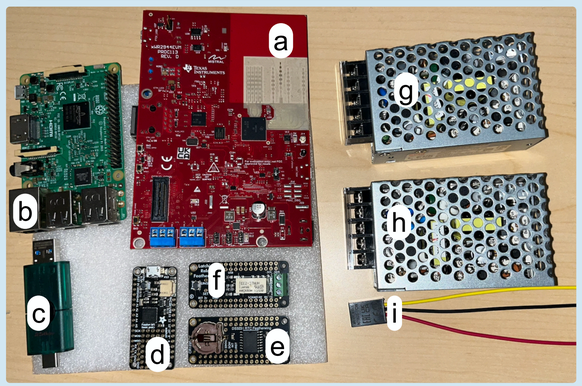


Fig. 3: Overview of the mmWave radar system's hardware components



Fig. 4: Fully integrated radar system



Fig. 5: 3D printed parabolic radar reflector using conductive PLA for beam focusing

AWR2944EVM (Fig. 3-a)

- TI mmWave FMCW radar front end with on-chip DSP producing processed outputs (TLV packets)

Raspberry Pi (Fig. 3-b)

- Experiment Orchestrator: Boot-time automation to start/stop experiments, run the radar CLI, and coordinate collection sequences
- Data + logs management: write radar outputs to external 256GB storage **(Fig. 3-c)**, organize acquisition sessions, and manage program execution/status logs.
- Remote status + power saving: send periodic health/status updates and trigger safe shutdown / standby to support low-power duty cycling

Feather MCU (Fig. 3-d)

- Implements a deterministic supervisory state machine for autonomous operation (run / idle / recovery)
- Accurate timekeeping using FeatherWing RTC **(Fig. 3-e**) 1 Hz square-wave interrupts for scheduling and wake windows
- Relay-based power control **(Fig. 3-f)** to cycle the Raspberry Pi and radar for low-power idle periods and recover from stalls (power sequencing + watchdog-style resets)
- DC–DC step-down converters **(Fig. 3-g,h,i)** to provide stable power supply and help prevent voltage drops during switching events.

TI's Signal Processing Chain and Radar Configurations

- Mapped TI's mmWave processing stages relevant to atmospheric returns:
  - chirp/frame acquisition → FFT processing → range-Doppler products → Range vs Doppler heatmaps
- Focused on how the radar data cube is formed (ADC Samples × chirps × antennas) and how waveform/DSP configuration parameters affect:
  - sensitivity and coherent integration gain
  - range and velocity resolution
  - unambiguous measurement limits
  - future angle-of-arrival (AoA) capability
- Defined configuration sets optimized for atmospheric sensing rather than sparse, high-reflectivity vehicle targets:
  - longer range / higher sensitivity
  - higher frame rate
  - higher resolution

DSP outputs (TLV extraction)

- Developed a custom Python pipeline to parse TI TLV packets and reconstruct:
  - Range–Doppler power matrices
  - Power-vs-Range slices across Doppler bins (beyond standard single-profile views)
  - Zero-doppler and noise profile
- Added validation checks to ensure consistency between radar-generated products before environmental analysis.

Noise reduction and weak-target enhancement

- Implemented multi-frame averaging methods to suppress noise fluctuations and improve weak-return visibility and consistency over time.
- Evaluated quantified trade-offs between range, velocity, and temporal resolution. The processing framework was designed to support future extensions including averaging across chirps, Doppler bins, and range bins, as well as clutter-aware filtering and thresholding methods.

RESULTS

External controller + automation reduced manual intervention and enabling:

- Reliable startup/config sequencing with minimized power consumption **(~5.3 W avg.)**
- Stable power management designed to minimize voltage fluctuations and support reliable recovery from system faults or stalled operation.
- Long-duration data collection with status visibility. At a rate of **1.37GB/day**, each radar system **(Fig. 6)** lasts up to 6 months.

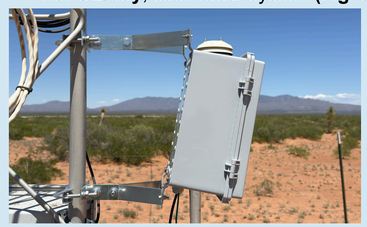


Fig. 6: Radar system #1 deployed at the Jornada Experimental Range, NM as of June 2, 2026.

Weak-return visibility / noise-floor stability via averaging

- Clearer object detection **(Fig. 7-a,b)**.
- Noticeably flatter noise floors at longer ranges **(Fig. 8-a,b)**.

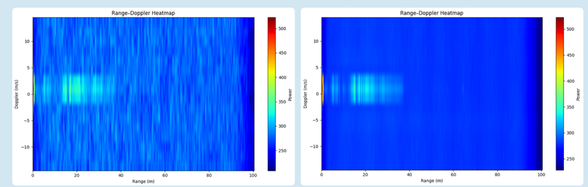


Fig. 7: Doppler vs. Range Heatmap of a tree at 20-40m without noise reduction **(a)** and with noise reduction (averaged over 150 frames) **(b)**

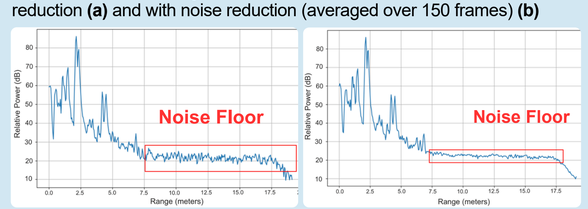


Fig. 8: Range Profile without noise reduction **(a)** and with noise reduction (averaged over 16 frames) **(b)**

Future work will focus on large-scale data acquisition and continued signal processing to assess atmospheric monitoring performance. Ten radar units are scheduled for deployment at the Jornada Experimental Range, NM to observe dust storms and summer monsoon rainfall.

ACKNOWLEDGMENTS

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